Effects of soil drainage types on macrofaunal community composition and density

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Abstract
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Keywords
soil macrofauna, soil drainage, Paracou, Amazon forest, soil ecology

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Table of contents

1 Introduction **1**
1.1 Context 1 . . . 1
1.2 Context 2 . . . 1
2 Material and Methods **1**
2.1 Study Site . . . 1
2.2 Sampling and Identification . . . 1
2.3 Statistical analysis . . . 1
Explorative Analysis ▪ Analysis of Abundances (Generalized Linear Mixed Models) ▪ Analysis of Community Composition (PERMANOVA)
3 Results **2**
3.1 General . . . 2
3.2 GLMM . . . 2
Global Abundance ▪ Formicidae ▪ Isoptera ▪ Lumbricina Oligochaetes
3.3 PERMANOVA . . . 3
4 Discussion **4**
5 Conclusion **4**
6 Acknowledgements **4**
7 Supplementary Material **4**
References **4**

1. Introduction

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1.1 Context 1

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1.2 Context 2

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2. Material and Methods

2.1 Study Site

Refer to Figure 1

2.2 Sampling and Identification

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2.3 Statistical analysis

Analysis was performed using R version 4.6.1 (R Core Team, 2026).

Explorative Analysis

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Analysis of Abundances (Generalized Linear Mixed Models)

glmmTMB (M. Brooks et al., 2026; M. E. Brooks et al., 2017) for fitting GLMMs negative binomial (NB2) and Poisson tested

Model diagnostics evaluated using DHARMa (Hartig, 2026)

Type II Wald χ^2 tests Anova() from car (Fox et al., 2026; Fox & Weisberg, 2019)

Analysis of Community Composition (PERMANOVA)

vegan (Oksanen et al., 2026): adonis2, betadisper, permutest, envfit permute (Simpson, 2026): permutation schemes

tidyverse (Wickham et al., 2019; Wickham, 2023) used for plotting and data processing

Spatial representation of sampling sites: in QGIS (QGIS Development Team, 2024). Monolith locations reconstructed using field measurements and reference trees from Paracou inventory data (Derroire et al., 2026a, 2026b) in R (R Core Team, 2026) and sf package (Pebesma, 2018; Pebesma & Bivand, 2023)

Estimated marginal means and Tukey pairwise comparisons using emmeans (Lenth & Piaskowski,

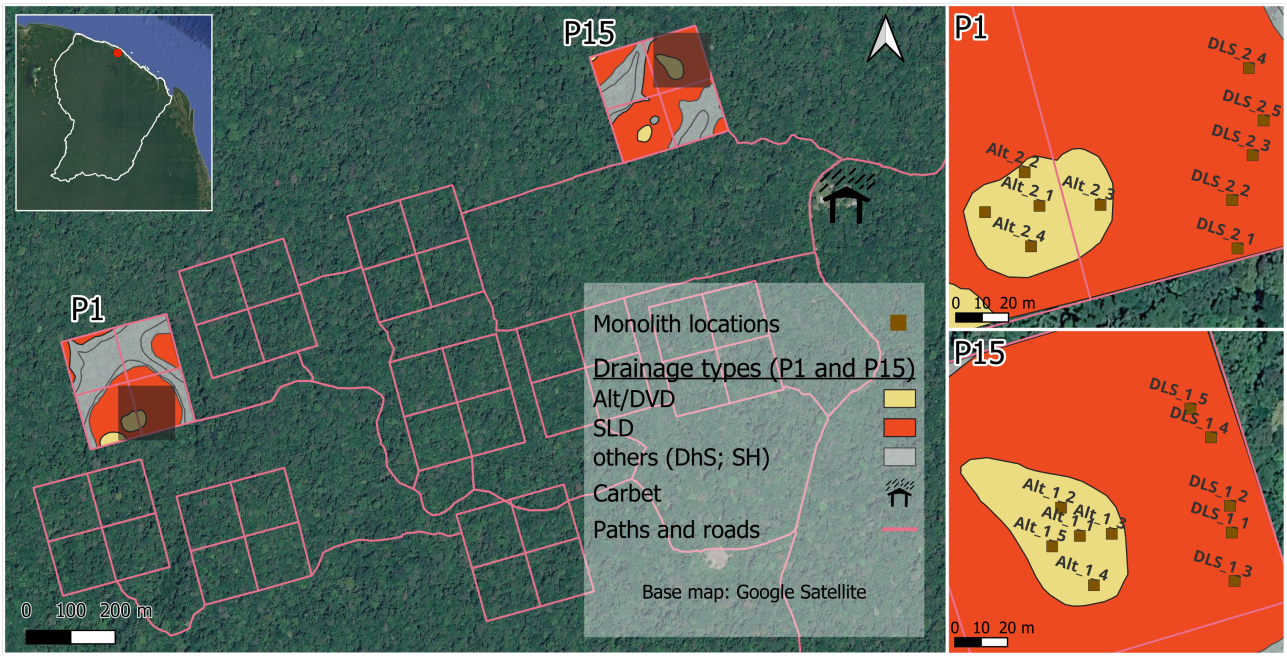


Figure 1. Overview map of Paracou with the sampling locations. Location of right panels shown by grey squares. White lines on the right panels represent altitude isolines.

2026).

3. Results

3.1 General

TEXT Densities by soil type and layer Figure 2

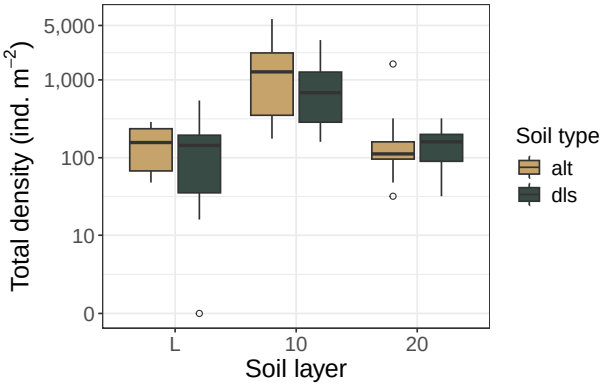


Figure 2. Boxplot with total macrofauna densities by soil type and layer. y-axis is scaled logarithmically.

TEXT Relative composition of community abundances Figure 3

TEXT Explorative ordination PCA Figure 4

3.2 GLMM

Effects of soil type and layer were tested on global, ant, termite and earthworm abundances. All retained models described below are listed in Table 2.

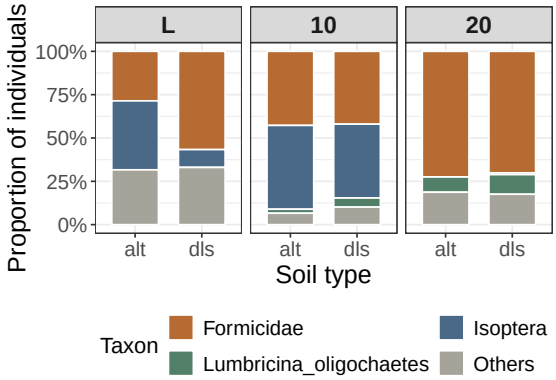


Figure 3. Relative composition of macrofaunal community abundances by soil type and layer.

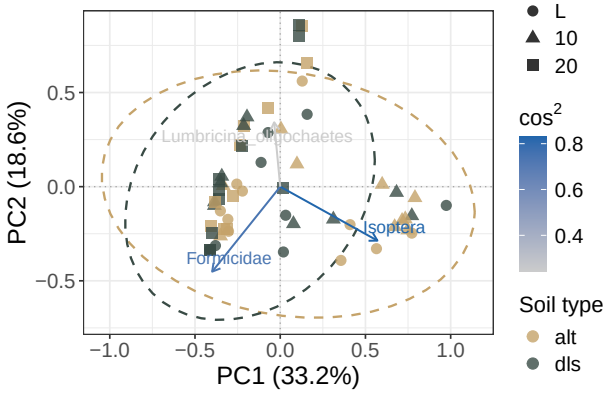


Figure 4. PCA ordination based on Hellinger-transformed community matrix. First two dimensions are presented. Only taxa with $\cos^2 > 0.1$ are presented.

Global Abundance

Type II Wald χ^2 tests revealed a highly significant effect of soil layer on total soil macrofauna abundance ($\chi^2 = 106.85$, df = 2, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.52$, df = 1, p = 0.469). Estimated random-effect variance (monolith) was 0.437 (SD = 0.661), indicating variation in baseline macrofauna abundance among monoliths.

Pairwise comparisons with Tukey adjustment (Table 1) showed that macrofauna abundance was significantly higher in layer 10 than in layers L (7.42 times) and 20 (6.18 times), while no significant difference was found between layers L and 20 (p = 0.714).

Formicidae

Type II Wald χ^2 tests revealed a significant effect of soil layer on ant abundance ($\chi^2 = 28.00$, df = 2, $p < 0.001$). The effect of soil type was not significant ($\chi^2 = 0.48$, df = 1, p = 0.486). The estimated random-effect variance for monolith was 0.655 (SD = 0.810), indicating variation in baseline ant abundance among monoliths. Quantile deviation was detected in the DHARMa residuals for the presented model. Specifically, upper quantile deviated significantly and thus relatively high-abundance observations may not be adequately captured by the model. Since the diagnostics did not indicate deviations regarding dispersion, zero-inflation, uniformity as well outliers, and the model has the best AIC support, it was retained, regardless.

Pairwise comparisons with Tukey adjustment (Table 1) showed that ant abundance was significantly higher in layer 10 than in layer L (5.05 times) and layer 20 (3.27 times), while no significant difference wasfound between layers L and 20 (p = 0.3995).

Isoptera

Type II Wald χ^2 tests revealed a highly significant effect of soil layer on termite abundance ($\chi^2 = 36.56$, df = 2, $p < 0.001$).The effect of soil type was not significant ($\chi^2 = 0.89$, df = 1, p = 0.347). The estimated random-effect variance for monolith was effectively zero (8.22×10^{-9}; SD = 0.00009), indicating virtually no remaining variation in baseline termite abundance among monoliths after accounting for the fixed effects.

Pairwise comparisons with Tukey adjustment (Table 1) showed that termite abundance was significantly higher in layer 10 than in layer L (18.08 times) and layer 20 (745.48 times; estimated richness for 20 nearing zero). Abundance was also significantly higher in layer L than in layer 20 (41.23 times; all p 0.0073).

Lumbricina Oligochaetes

Type II Wald χ^2 tests indicated a significant effect of soil layer on earthworm abundance ($\chi^2 = 12.69$), df = 2, p = 0.0018),whereas soil type had no significant effect ($\chi^2 = 0.35$), df = 1, p = 0.557). Pairwise comparisons showed significantly higher earthworm abundance in layer 10 compared to layer 20 (2.31 times higher). No significant differences were detected between layer L and the other layers. However, no earthworms were recorded in layer L, resulting in unstable model estimates and unbounded confidence intervals for this layer. The Estimated random-effect variance for monoliths was 0.510 (SD = 0.714)

3.3 PERMANOVA

PERMANOVA did not show a significant difference in macrofaunal community composition between soil types (F = 1.68, $R^2 = 0.085$, p = 0.139). The test for homogeneity of multivariate dispersion was also non-significant (F = 0.11, p = 0.742), indicating no detectable difference in within-group dispersion between soil types. Layers were aggregated per monolith to account for spatial dependence.

PERMANOVA revealed a significant effect of soil layer on macrofauna community composition (Bray–Curtis dissimilarity: F = 5.46, $R^2 = 0.163$,p = 0.0001). Soil layer explained around 16.3% of variation in the community composition, indicating that macrofauna community composition differed significantly among layers. Permutation was restricted within monolith to account for spatial dependence of layers. Homogeneity of multivariate dispersion was assessed using a permutation test based on betadisper. No significant differences in dispersion were detected for soil layers (F = 0.75, p = 0.321).

Pairwise PERMANOVA suggests significant differences in macrofauna community composition for all soil layers (Table 3). Significant differences were observed between L and 10 (F = 6.05, $R^2 = 0.141$, adjusted p = 0.0003), L and 20 (F = 4.55, $R^2 = 0.110$, adjusted p = 0.0003), and 10 and 20 (F = 5.76, $R^2 = 0.132$, adjusted p = 0.0012), using Bonferroni correction.

Table 1. Pairwise emmeans comparisons of abundance by layer for total macrofauna, ants (Formicidae) and termites (Isoptera).

Taxon	Comparison	Abundance ratio	z-ratio	Adjusted p-value
Total macrofauna	L - 10	0.13	-9.12	<0.0001***
Total macrofauna	L - 20	0.83	-0.78	0.713689
Total macrofauna	10 - 20	6.18	8.34	<0.0001***
Ants (Formicidae)	L - 10	0.20	-4.96	<0.0001***
Ants (Formicidae)	L - 20	0.65	-1.29	0.399482
Ants (Formicidae)	10 - 20	3.27	3.84	0.000363***
Termites (Isoptera)	L - 10	0.06	-4.06	0.000145***
Termites (Isoptera)	L - 20	41.23	3.01	0.007262**
Termites (Isoptera)	10 - 20	745.48	5.42	<0.0001***

Abundance ratios are exponentiated log-scale contrasts. Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001. P-values adjusted using the Tukey method within each taxon.

Table 2. Summary of retained GLMMs for global abundance, ants (Formicidae), termites (Isoptera) and earthworms (Lumbricina oligochaetes). ADD MORE DESCRIPTION HERE

Response	Distribution	Predictor	chi_square	df	p-value
Total macrofauna	Negative binomial	soil_type	0.52	1	0.46933
Total macrofauna	Negative binomial	layer	106.85	2	<0.0001***
Ants	Negative binomial	soil_type	0.48	1	0.48633
Ants	Negative binomial	layer	28.00	2	<0.0001***
Termites	Negative binomial	soil_type	0.89	1	0.34664
Termites	Negative binomial	layer	36.56	2	<0.0001***
Earthworms	Poisson	soil_type	0.35	1	0.55692
Earthworms	Poisson	layer	12.69	2	0.00176**

Type II Wald Chi-square tests. All models included soil type and soil layer as fixed effects and monolith ID as a random intercept. Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001

Table 3. Pairwise PERMANOVA for community composition comparisons between layers.

Comparison	F-statistic	R^2	Adjusted p-value
L - 10	6.05	0.141	0.0003***
L - 20	4.55	0.110	0.0003***
10 - 20	5.76	0.132	0.0021**

Significance levels: * p < 0.05; ** p < 0.01; *** p < 0.001; p-values adjusted using Bonferroni method

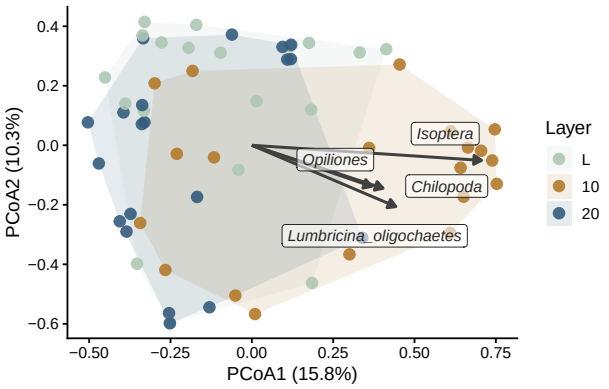


Figure 5. Bray-Curtis PCoA CAPTION HERE

4. Discussion

TEXT - no differences in densities or community composition for soil types (argue about same vegetation cover etc.) - clear differences in both for layers (connect to literature knowledge)

5. Conclusion

TEXT

6. Acknowledgements

7. Supplementary Material

Supplementary material, including detailed statistical analysis, scripts, data, QGIS layers, additional software attribution as well as the Quarto file used to produce this article, is available from url={https://github.com/jasonberger2002/FTH_2026_soil_macrofauna}.

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